Galois Groups of Trinomials with Small Coefficients

Hilel Garmi

September 10, 2026

By Hilbert's irreducibility theorem [Hil92], almost all rational polynomials of degree n has Galois group S_n over $\mathbb{Q}$. Nevertheless, giving explicit examples of such polynomials for all n remained a difficult problem. Only 38 years after Hilbert's work, Schur [Sch31] showed that the Laguerre polynomials

$$L_n^{(0)} := \sum_{k=0}^{n} \binom{n}{k} \frac{x^k}{k!}$$

have Galois group S_n for every n. Later, the same was proven for other families of generalized Laguerre polynomials, such as the Hermite polynomials, $L_n^{(\pm 1/2)}$ [Sch31], and the reverse Bessel polynomials, $L_n^{(-2n-1)}$ [Gro78] [FT02]. However, it appears that the only known example with a bounded height is the family Selmer trinomials $f_n(x) = x^n - x - 1$ which was shown to be irreducible by Selmer [Sel56] and to have Galois group S_n by Nart and Vila [NV79]. To achieve that, they first showed that all the inertia groups of f_n are either trivial or generated by a transposition. They then utilized Hermite's Theorem to prove that the Galois group of f_n over $\mathbb{Q}$ is generated by its inertia groups, and deduced that it is S_n since any transitive permutation group generated by transpositions is S_n. It is natural to try to apply a similar strategy for $f_n(x) = x^n + ax + b$, or its irreducible factors, for some other values of $a, b \in \mathbb{Z}$.

Nart and Vila [NV79] themselves and later Osada [Osa87] indeed generalized the above method to obtain some sufficient criteria on a, b and n for the Galois group of f_n to be S_n, assuming its irreducibility. Yet, each of these criteria includes some congruence condition on n, leaving out a positive density of natural numbers as exceptions. Maybe the most instructive example for how these congruence conditions arise is the reducible famiy $f_n = x^n - 2x + 1$, which prompted our own interest in the subject due to a recent result of Bary-Soroker, Shusterman, and Zannier [HW17, Appendix A]. In this case, f_n factors as the product of $x-1$ and the polynomial $g(x) = x^{n-1}+x^{n-2}+\cdots+x-1$. The latter polynomial was shown to be irreducible by Miles [Mil60] and Wolfram [Wol98], which were in fact interested in its close relative, the generalized Fibonacci polynomial $x^{n-1} - x^{n-2} - \cdots - x - 1$. Martin [Mar04] succeded, using methods similar to those used in the classical case of $x^n - x - 1$, to show that the Galois group of g is S_{n-1} for any odd value of n. The reason his proof fails for even values of n, is that in this case

$$x^n - 2x + 1 \equiv x^n - 1 \equiv \left(x^{n/2} - 1\right)^2 \mod 2$$

and thus the inertia group of f_n over $\mathbb{Q}_2$ is no longer generated by a transposition.

In this talk I will present a proof that bypasses this problem. The previous arguments, both for the cases of generalized Laguerre polynomials, and for the case of $x^n - x - 1$, focused on local methods, which are insufficient for trinomials with bigger coefficients. Our new approach utilizes global bounds on the discriminant of number fields, like the Minkowski's bound and the stronger Odlyzko's bound [Odl77], to show that the "unproblematic" inertia groups, those that are indeed generated by transpositions, already generate the entire Galois group of f_n, rendering the problematic ones dispensable. Using this approach, for some small values of $a, b \in \mathbb{Z}$, and in particular in any case where $a, b \in \{\pm 1, \pm 2\}$, and for any $n \in \mathbb{N}$, we were able to determine the factorization of f_n and show that the Galois group of each of

its irreducible factors, is indeed symmetric.

References

[FT02] Michael Filaseta and Ognian Trifonov. The irreducibility of the Bessel polynomials. *Journal für die reine und angewandte Mathematik*, 550:125–140, 2002.

[Gro78] Emil Grosswald. *Bessel Polynomials*, volume 698 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, Heidelberg, 1978.

[Hil92] David Hilbert. Ueber die irreducibilität ganzer rationaler Functionen mit ganzzahligen Coefficienten. *Journal für die reine und angewandte Mathematik*, 1892(110):104–129, 1892.

[HW17] W. Patrick Hooper and Barak Weiss. Rel leaves of the Arnoux–Yoccoz surfaces. *Selecta Mathematica*, 24(2):875–934, December 2017.

[Mar04] Paulo A Martin. The Galois group of $x^n - x^{n-1} - \cdots - x - 1$. *Journal of Pure and Applied Algebra*, 190(1):213–223, 2004.

[Mil60] E. P. Miles. Generalized Fibonacci numbers and associated matrices. *The American Mathematical Monthly*, 67(8):745–752, 1960.

[NV79] E. Nart and N. Vila. Equations of the type $x^n + ax + b$ with absolute Galois group s_n. *Proceedings of the sixth conference of Portuguese and Spanish mathematicians*, pages 821–825, January 1979.

[Odl77] A. M. Odlyzko. Lower bounds for discriminants of number fields, II. *Tohoku Mathematical Journal*, 29(2):209 – 216, 1977.

[Osa87] Hiroyuki Osada. The Galois groups of the polynomials $x^n + ax^1 + b$. *Journal of Number Theory*, 25:230–238, 1987.

[Sch31] Issai Schur. Affectless equations in the theory of laguerresches and hermitian polynomials. *Journal für die reine und angewandte Mathematik*, 165:52–58, 1931.

[Sel56] Ernst S. Selmer. On the irreducibility of certain trinomials. *Mathematica Scandinavica*, 4(2):287–302, 1956.

[Wol98] D. A. Wolfram. Solving generalized Fibonacci recurrences. *The Fibonacci Quarterly*, 36(2), 1998.